AI Agent Setup for Shadow Network

Branching, agent instructions, and agent skills

Members	Busaco, Aude Mar Hyra V. Cabanting, Rwoen Dave O. Areglo, Brent E.
Project	Shadow Network
Working branch	codex/set-up-project-skills
Date	October 12, 2026

Overview

This document records the agent setup for Shadow Network, an Android-first Flutter app for offline SOS and chat messaging. It covers the working branch, project instructions, two custom skills, and two skills obtained from online providers. Each setup section includes a space for its supporting screenshot.

1 New Git branch	codex/set-up-project-skills
2 Agent instructions	AGENTS.md at the repository root
3 Custom skills	shadow-network-relay-architect shadow-network-validation-planner
4 Provider skills	flutter-expert (Jeff Allan) security-threat-model (OpenAI)

1 Create a New Git Branch

A dedicated working branch named codex/set-up-project-skills is checked out for the agent setup. Keeping this work on a separate branch allows the instruction file and skill additions to be reviewed before they are merged into main.

Verification commands

```
git branch --show-current
git branch -vv
git status --short
```

The verified branch name is codex/set-up-project-skills. At the time of documentation, the branch has no remote tracking entry, and the AGENTS.md update and .agents folder are local, uncommitted changes.

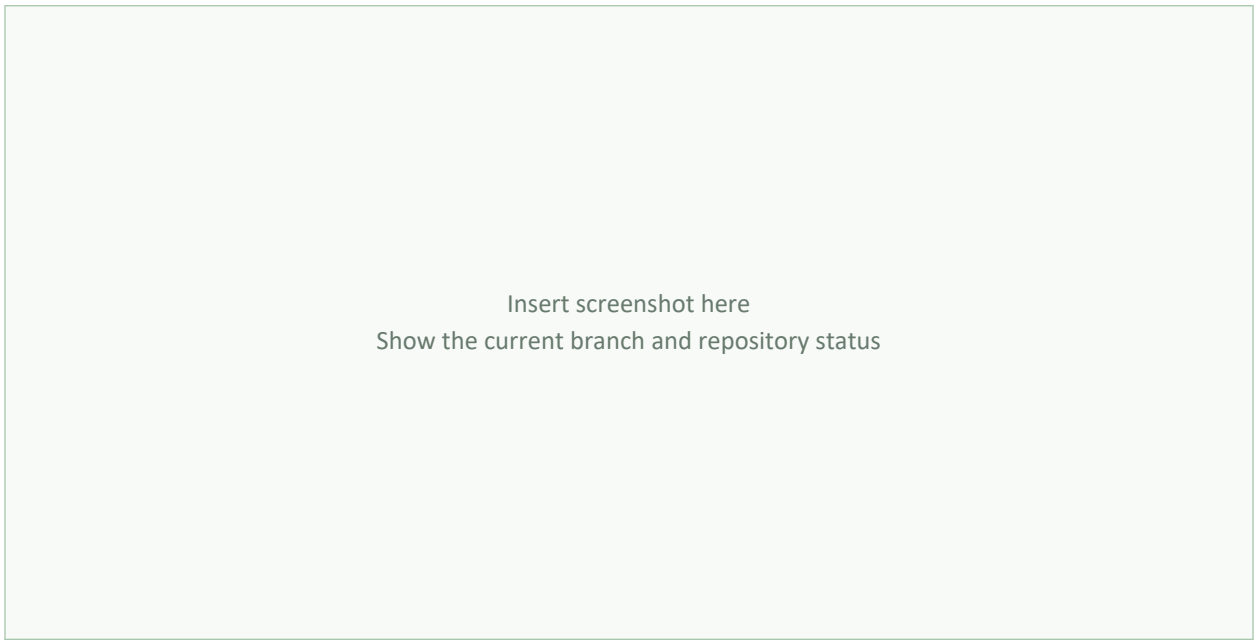


Figure 1. The Shadow Network setup branch checked out locally.

2 Create an Agent Instructions File

AGENTS.md is located at the repository root. We updated the existing guide to give coding agents the project context and identify the skills available for recurring development tasks. It covers the following areas:

- **Project structure:** The Flutter app runs from the root. Shared code is in lib/app and lib/core, while features use data, domain, and presentation layers.
- **Coding patterns:** Widgets focus on rendering, Riverpod manages state, and repositories separate business rules from SQLite and platform implementations.
- **Development commands:** The guide lists flutter pub get, flutter analyze, flutter test, flutter run, and flutter build apk.
- **Verification:** Focused tests cover payloads, deduplication, expiry, hop counts, migrations, encryption, trust, and validation metrics.
- **Offline and security rules:** Emergency messaging and maps must remain usable offline. Agents must preserve trust rules and avoid exposing secrets.
- **Skill sources:** The guide identifies both custom skills and both provider skills, including links to their pinned source revisions.

These instructions keep agent work aligned with the existing Android-first app, its Riverpod architecture, and its offline communication requirements.

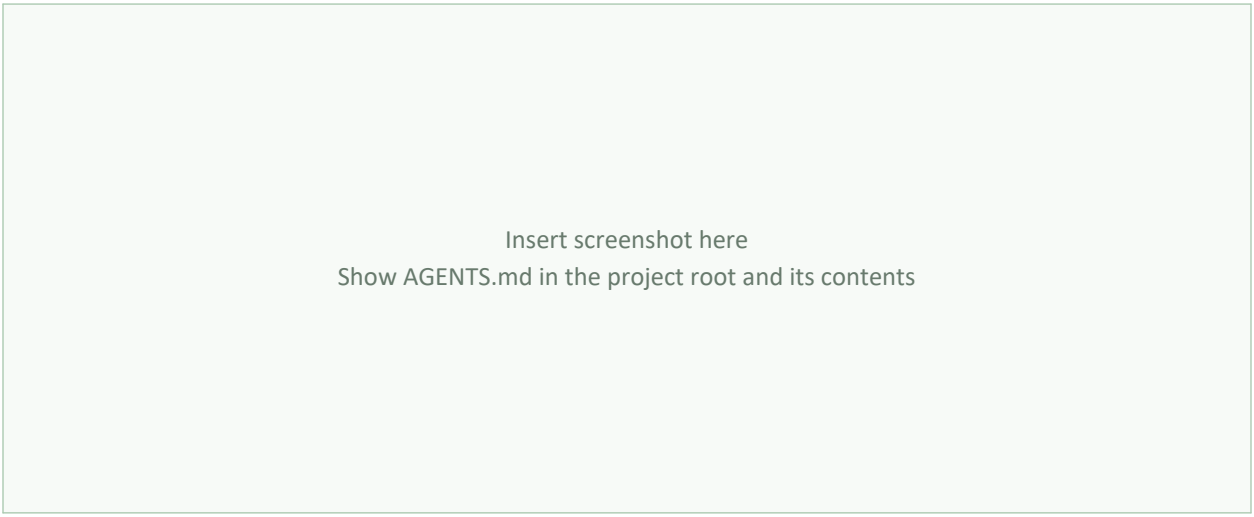


Figure 2. AGENTS.md in the Shadow Network repository root.

3 Create Two Custom Skills

We created two skills for recurring Shadow Network tasks. Each has its own folder under .agents/skills and a SKILL.md file containing its name, description, and project-specific instructions.

Shadow Network Relay Architect

`shadow-network-relay-architect`

Purpose: Implement or review offline SOS and chat relaying, SQLite queues, BLE and Wi-Fi Direct transport, and Store-Carry-Forward behavior.

- **Relevant files:** The skill points to payload codecs, SCF storage and relay services, the transport interface, the Android bridge, and runtime providers.
- **Relay contracts:** It preserves message hashes, deduplication, TTL expiry, hop tracking, and per-peer delivery state. Retries must not extend expiry.
- **Security boundaries:** Encrypted targeted bodies remain unreadable to relay-only devices, and trust or revocation rules apply to received and outbound messages.
- **Verification:** It uses focused relay, payload, encryption, and trust tests, while keeping physical Android checks separate from mock results.

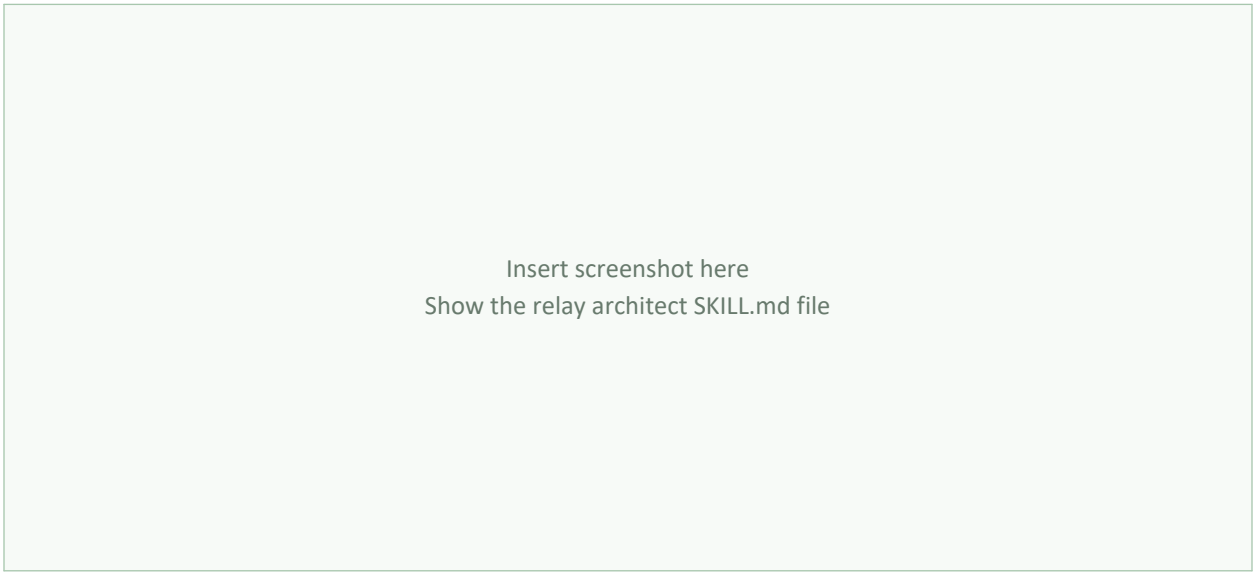


Figure 3. The shadow-network-relay-architect custom skill.

Shadow Network Validation Planner

`shadow-network-validation-planner`

Purpose: Plan repeatable field trials and interpret JSON or CSV session evidence for discovery, delivery, latency, hops, transport fallback, and battery impact.

- **Evidence pipeline:** The skill identifies validation events, the metrics calculator, export service, SQLite repository, and validation tests.
- **Repeatable trials:** Plans record device roles, network topology, Android versions, radio settings, message workload, distance, duration, and expected results.
- **Metric definitions:** It distinguishes successful peer transfers from end-to-end recipient delivery and separates node latency from propagation time.
- **Reporting quality:** Raw records stay intact. Missing measurements remain unavailable, and simulated, measured, and proposed values are labeled separately.

This skill supports the app's validation module and helps the group explain what its recorded metrics actually demonstrate, without inventing measurements or overstating mock test results.

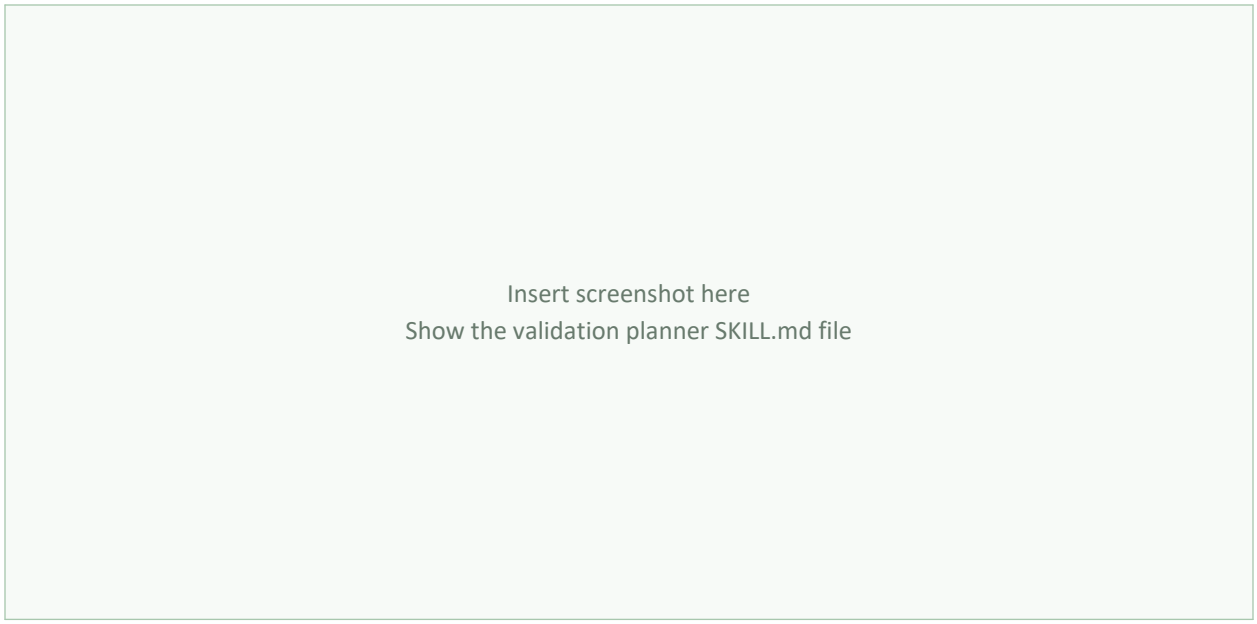


Figure 4. The shadow-network-validation-planner custom skill.

4 Install Skills from a Provider

Two published skills were downloaded into .agents/skills alongside the custom skills. Their original instructions, reference files, and licenses were retained. Both support work on the application itself.

flutter-expert	Jeff Allan / Synergetic Solutions. Guides Flutter widget development, Riverpod state, platform integration, widget testing, and performance. Applied within the app's existing architecture.
security-threat-model	OpenAI. When threat modeling is explicitly requested, identifies assets, trust boundaries, abuse paths, and mitigations. Relevant to peer payloads, device identity, encrypted messages, trust bundles, and registration.

Flutter source: Jeffallan/claude-skills (revision 1be15d8064f8)

Security source: openai/skills (revision 49f948faa925)

All four skills passed the skill-format validator. The downloaded provider files were also checked against their pinned source revisions. These checks establish valid packaging and provenance; they do not replace app tests.

Insert screenshot here
Show all four folders under .agents/skills

Figure 5. The two custom and two provider skills installed in the project.

Summary

Shadow Network now has a dedicated working branch, an updated AGENTS.md guide, and four project-local skills. The custom skills cover offline relay development and field validation. The provider skills add Flutter implementation guidance and security threat modeling.